PHY101: Introduction to Physics-1

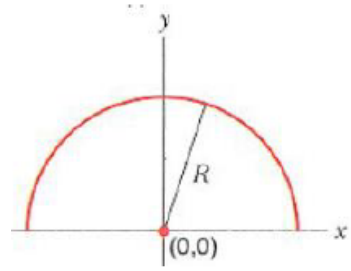
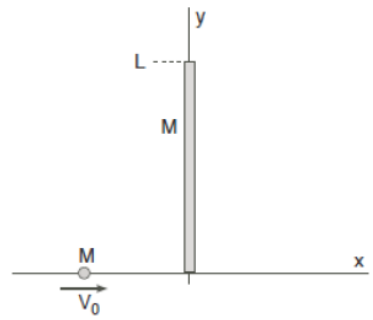
Monsoon, 2017

Tutorial-12

1. A disc of radius *R is* cut out from a larger disc of radius 2R in such a way that the edge of the hole touches the edge of the disc. Locate the centre of mass of the residual disc.

2. Dr. X (50 kg) and Dr. Y (60 kg) are seated at the two extremes of a 4 m long boat with mass 40 kg standing still in water. To discuss a research problem, they come to the middle of the boat. How far does the boat move on the water during this process? Neglect the friction of boat with water.

3. A wire is bent into a semicircular shape of radius *R* as shown in the figure below. The density of the wire varies as $\rho(x, y) = \rho_0(1 + ax + by)$, where ρ_0, a and b are the constants of appropriate dimension. Calculate the centre of mass (X,Y) of the wire. What happens in the limit $a \rightarrow 0$ and $b \rightarrow 0$?



4. A slender uniform rod of length *L* and mass *M* rests along the *y*-axis on a frictionless, horizontal table. A particle of equal mass *M* is moving along the *x*-axis at a speed V_0. At t=0 the particle strikes the end of the rod and sticks to it. Find the position $\mathbf{R}_{cm}(t)$ and velocity $\mathbf{V}_{cm}(t)$ of the center of mass of the system as a function of time.